



RUB

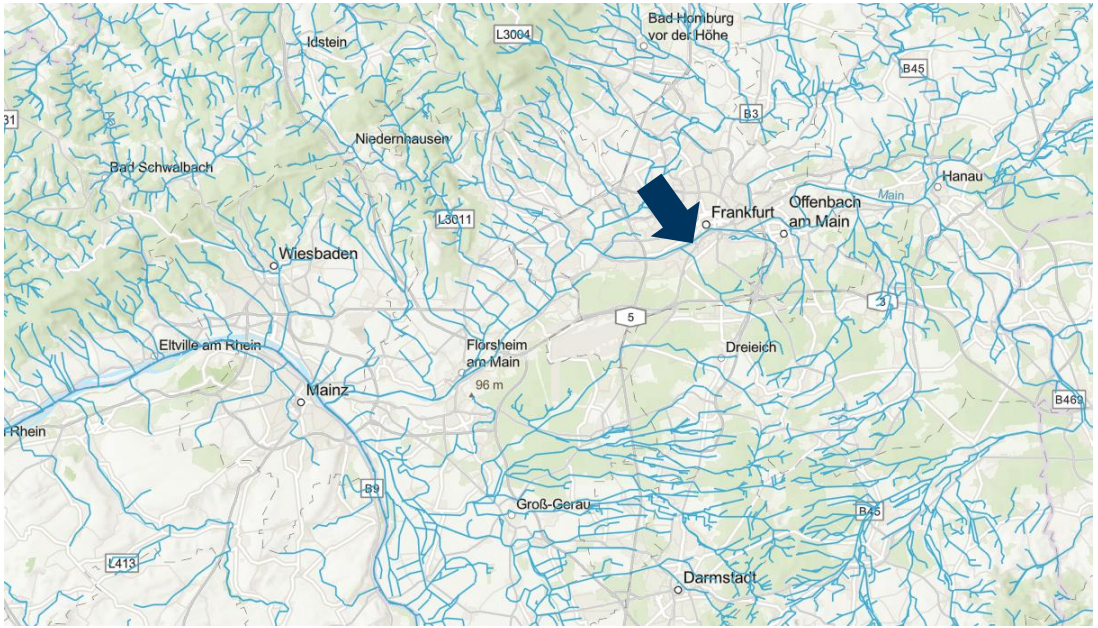
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EXERCICE WATER QUALITY

Monitoring data from the German River Main – Katrin Schweden

Monitoring Station River Main





Monitoring Data

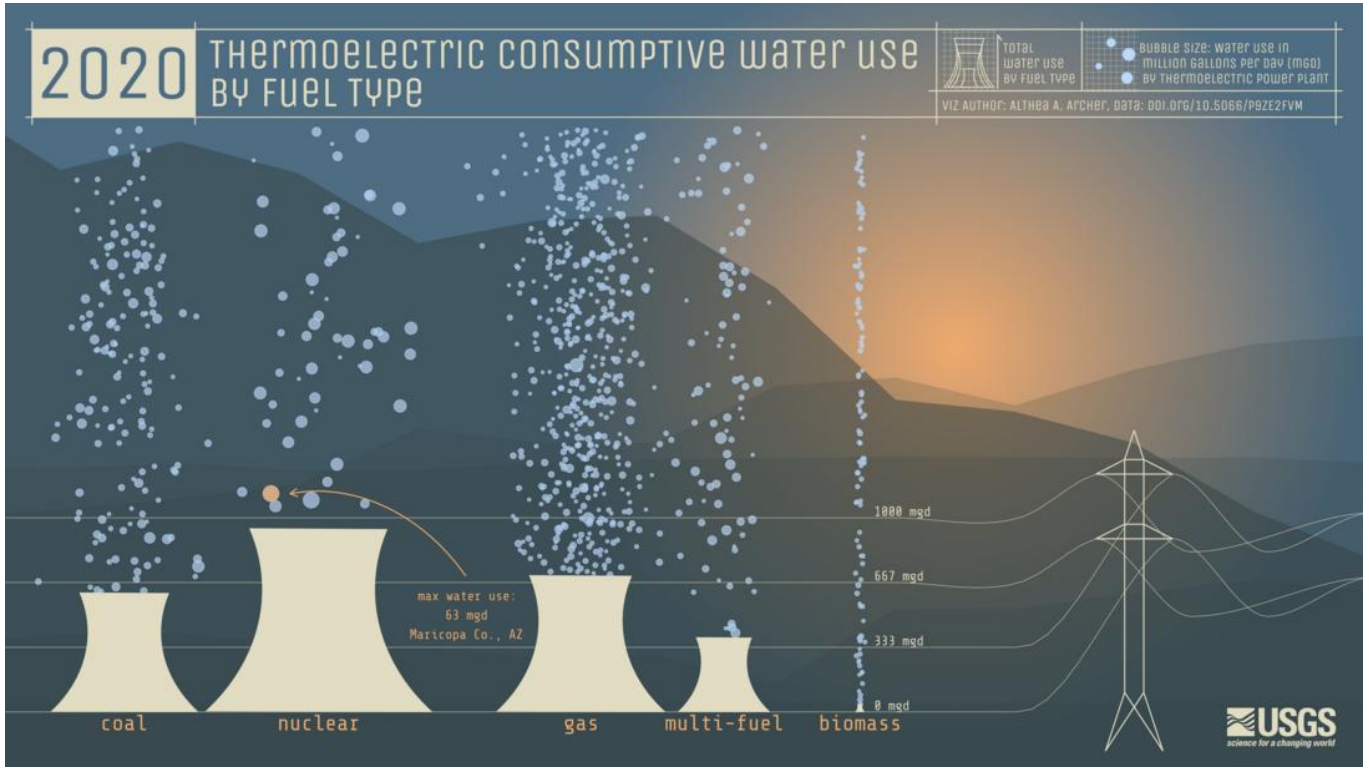
- **Electrical Conductivity in µS/cm (EC)**
- **Water Temperature in °C (Temp)**
- **Dissolved Oxygen in mg/l (DO)**
- **pH – Value**

Start date: 17.01.2007

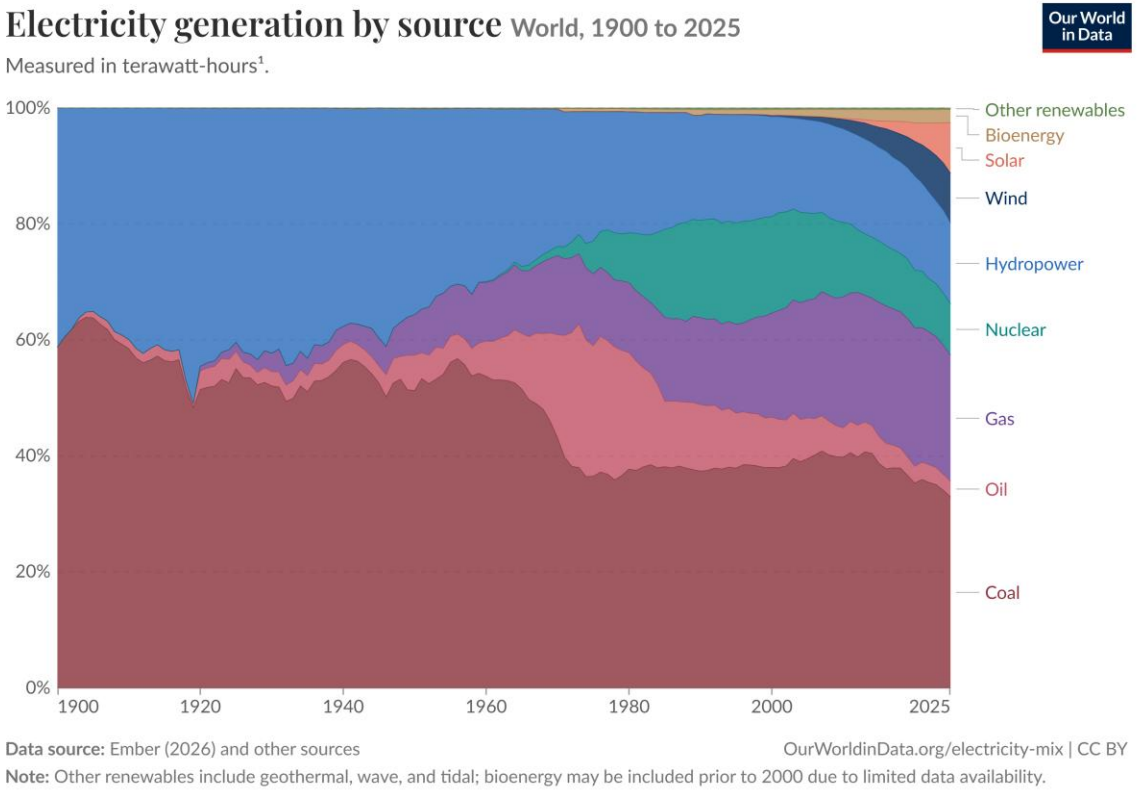
End date: 27.08.2026

Temporal Resolution: 0,5 h

Water Temperature



Water Temperature



Water Temperature



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Jul 2020

ARTICLE

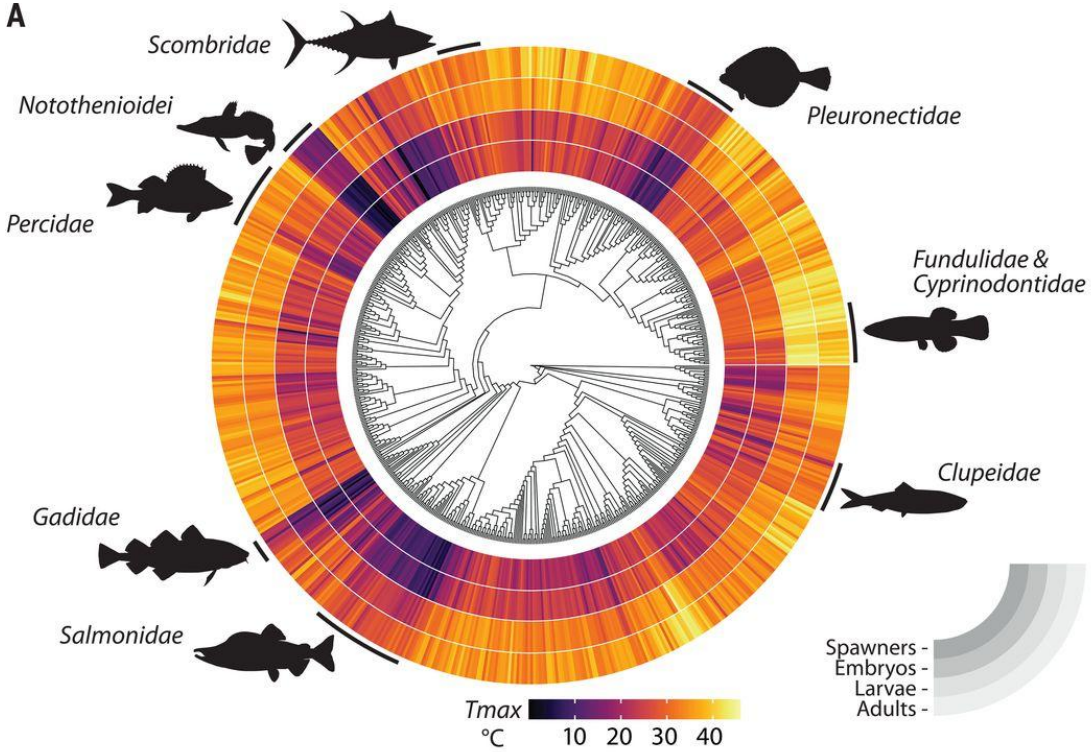
Thermal bottlenecks in the life cycle define climate vulnerability of fish

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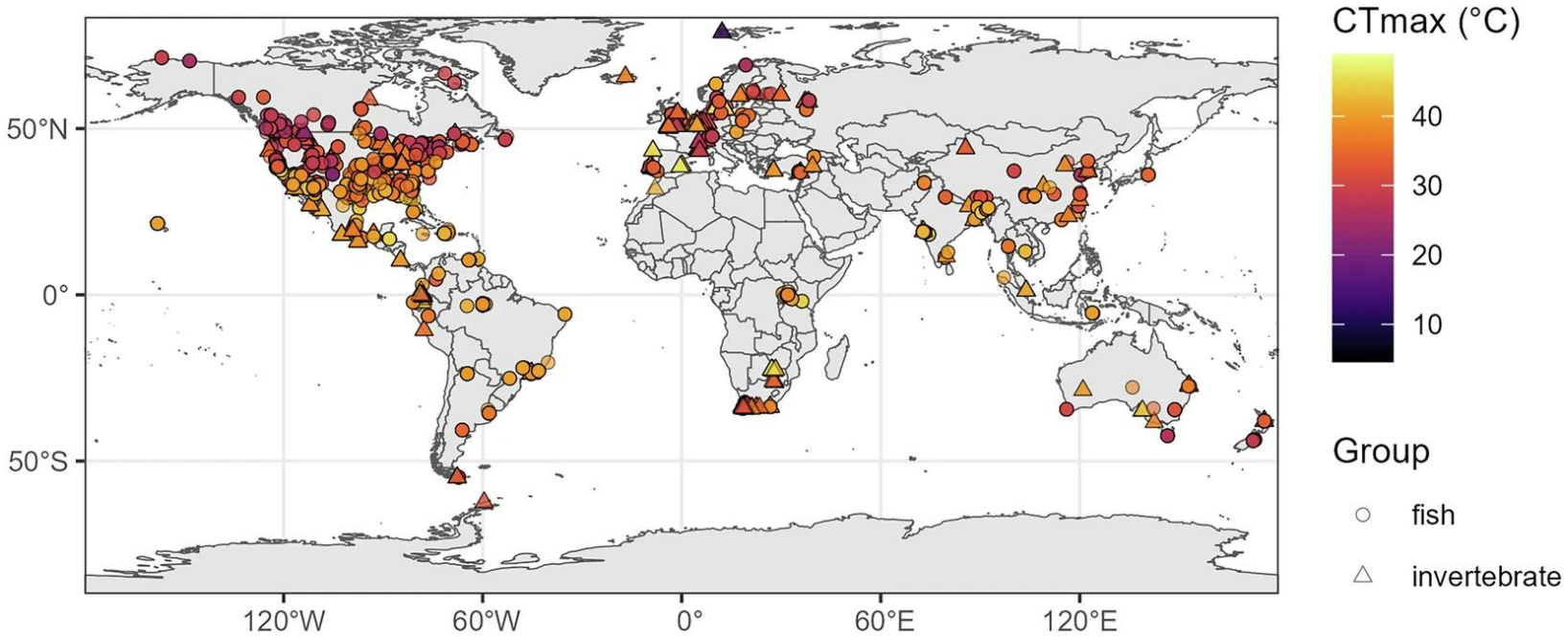
Flemming T. Dahlke, Sylke Wohlrab, Martin Butzin, and Hans-Otto Pörtner

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https://doi.org/10.1126/science.aaz3658



Water Temperature



Bayat, H.S., He, F., Medina Madariaga, G. et al. Global thermal tolerance compilation for freshwater invertebrates and fish. Sci Data 12, 1488 (2025). https://doi.org/10.1038/s41597-025-05832-w

Dissolved Oxygen

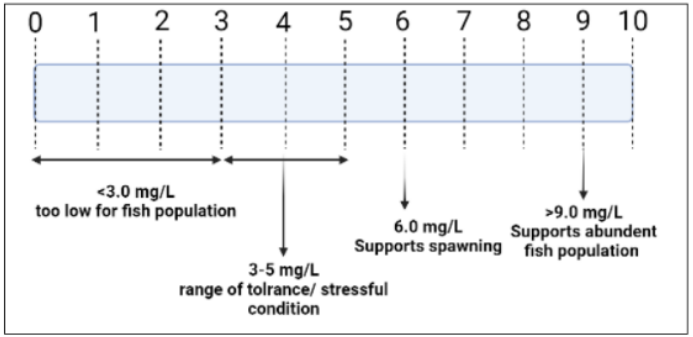
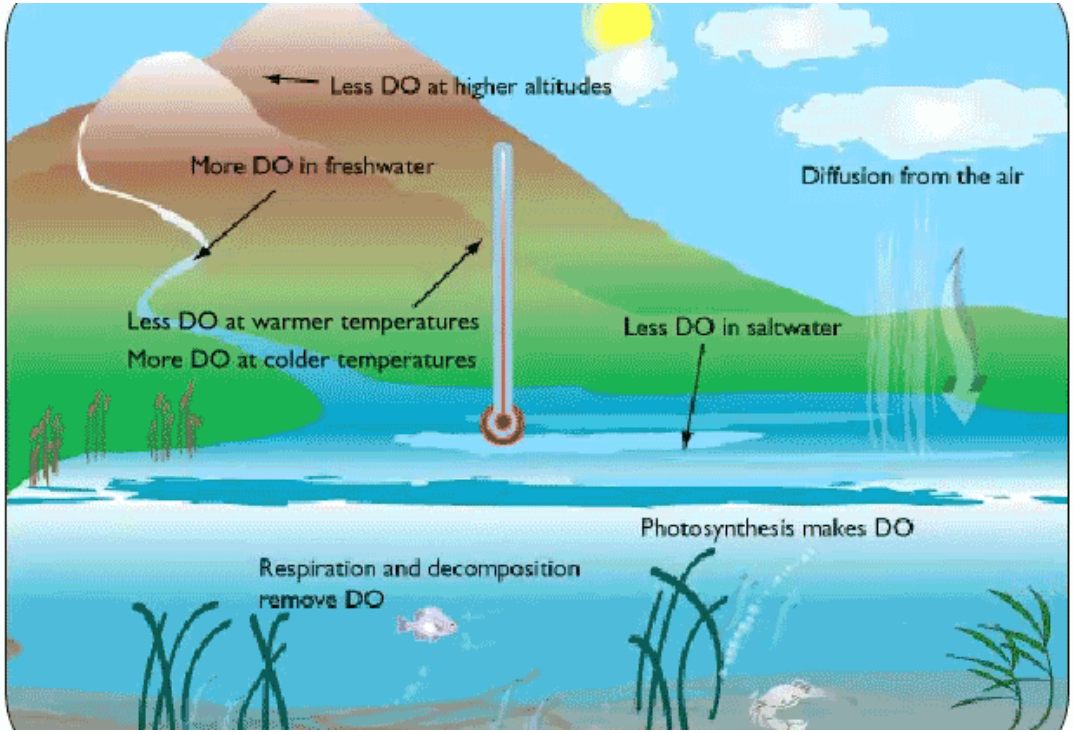


Fig 1: Range of tolerance for dissolved oxygen in fish

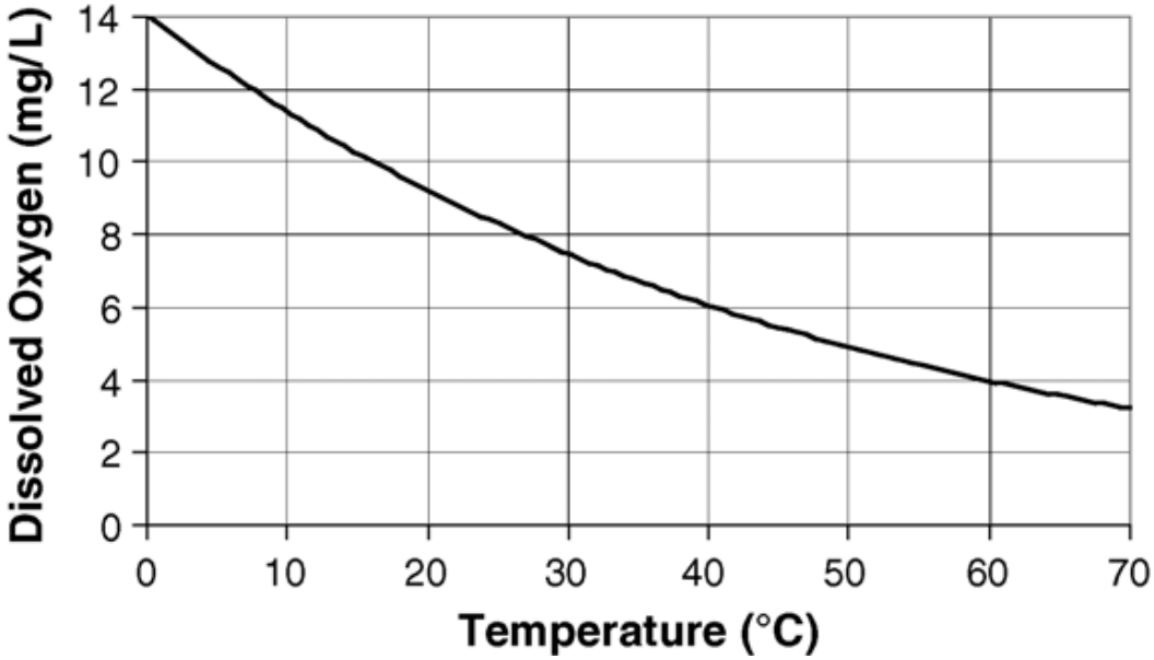
Ali, Bulbul & ., Anushka & Mishra, Abha. (2022). Effects of dissolved oxygen concentration on freshwater fish: A review. International Journal of Fisheries and Aquatic Studies. 10. 113-127. 10.22271/fish.2022.v10.i4b.2693.



https://coast.noaa.gov/estuaries/science-data/

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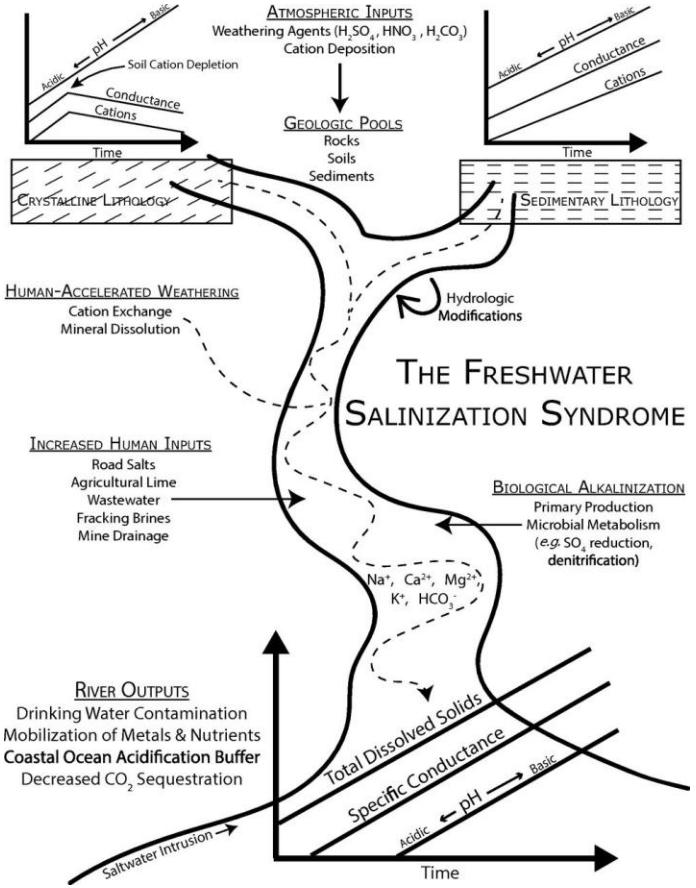
Dissolved Oxygen



Effect of temperature on oxygen solubility in water (generated by extrapolation of data from Vesilind, 1996).

Salinity – Electrical Conductivity

- Water with an EC of ~ 700 µS/cm is not suitable for irrigation.
- No reduction of salinity by decay only by dilution



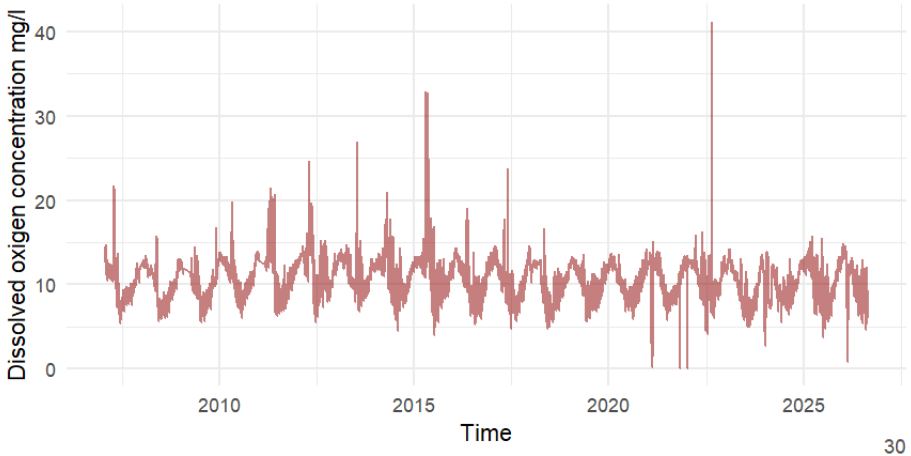
Exercise

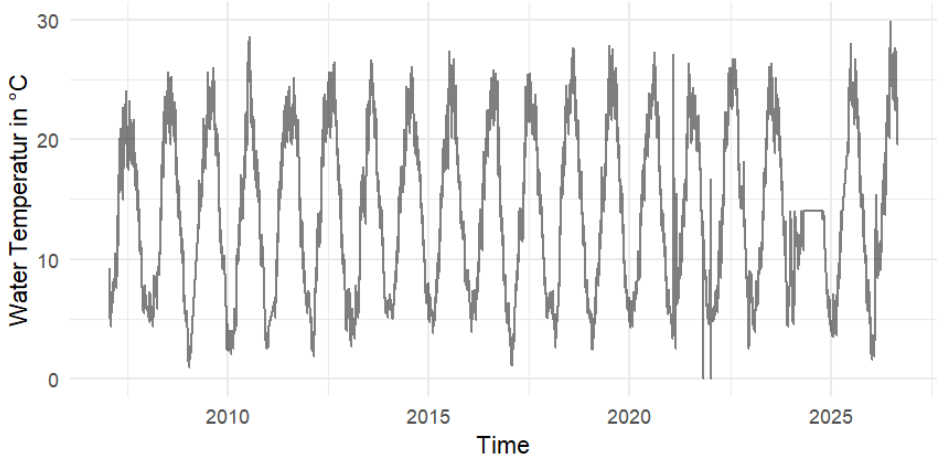
General data analyses. Try to identify:

- **Maxima, minima, mean, average, trends….**
- **Seasonalities (periodical changes within a year)**
- **Dependencies of the different parameters**
- **Check EC and DO for the given thresholds**
- **Pattern in decrease and increase of measured values**

Additional analysis

- **Reduce the temporal resolution of the data from 0.5 h to daily, weekly, or monthly.**
- **Do you see an effect on your results in comparison?**
- **Bonus: Can you find the current drought in the data?**


Dissolved oxigen concentration mg/l
40
30
20
10
0
2010
2015
2020
2025
Time


Water Temperatur in °C
30
20
10
0
2010
2015
2020
2025
Time

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